

FITMASTER AI / FITNESSMASTER

Complete Technical Documentation & System Specifications Manual

Platform: FitMaster AI Gym & Fitness System	Version: 2.5 High-Energy Edition
Backend Stack: Python 3.14 & Django 6.0	Database: SQLite3 (db.sqlite3)
Frontend: HTML5, CSS3, Bootstrap 5, Three.js	Date: August 2, 2026

1. Executive Summary & System Architecture

FitMaster AI (FitnessMaster) is a dynamic, high-energy web application engineered for comprehensive gym management, trainer client allocation, and intelligent customer workout and diet tracking. The platform operates on a robust three-tier architecture with full database persistence in SQLite3.

Layer	Technologies Used	Key Functional Responsibilities
Frontend UI	HTML5, CSS3 Vanilla, Bootstrap 5, FontAwesome 6	Responsive layouts, Cyber glassmorphism, 60:30:10 color system.
Interactive Graphics	Three.js 3D Visualizer, Chart.js Analytics	3D body muscle visualizer, revenue growth charts, macro doughnut graphs.
Backend Core	Python 3.14, Django 6.0, WhiteNoise Engine	MVC routing, authentication, role control, session security.
Database	SQLite3 (db.sqlite3) Relational DB	Persistent storage for users, plans, payments, products, logs, notices.

2. User Roles & System Capabilities

A. Administrator Role (Super Admin)

- Time-Based Package Builder:** Create & configure 3-month, 6-month, and 1-year membership packages with prices & descriptions.
- User & Role Management:** Edit customer and trainer details, activate/deactivate accounts, and adjust roles.
- Membership Dates Setting:** Manually configure custom start and end dates for customer membership packages.
- E-Commerce Product Center:** Manage supplement products, set prices, upload images, and control store inventory.
- Payments & Invoice Center:** View transaction history and generate downloadable/printable invoice receipts.
- Broadcast Email Notices:** Dispatch email announcements to customers, trainers, or all platform users.

B. Customer Role (Gym Member)

- Package Selection & Checkout:** Browse time-based membership plans and execute secure checkout.
- Assigned Exercise & Diet Tracking:** View personalized workout routines and daily meal plans created by their assigned trainer.

- **Monthly Weight, Height & BMI Tracking:** Log monthly body measurements with automated BMI calculations.
- **Supplement Store:** Purchase supplements and fitness equipment with stored order history.
- **Feedback & Review Submission:** Submit 1-5 star ratings and reviews for gym trainers and services.

C. Trainer Role (Certified Coach)

- **Category Specialization:** Select training category: **Gym Training**, **Yoga**, or **Zumba**.
- **Client Roster Allocation:** Add and assign allocated gym customers to personal training rosters.
- **Personalized Workout Routine Builder:** Create and assign custom workout exercises and duration targets.
- **Personalized Diet Plan Builder:** Create custom Breakfast, Lunch, and Dinner meal plans for clients.
- **Session Scheduling & Progress Logging:** Schedule 1-on-1 sessions and log progress metrics (weight/height/BMI/notes).

3. Database Schema & Entity Specifications

The platform defines 14 relational Django database models in users/models.py with 100% data persistence in SQLite3:

Model Name	Primary Fields	Description & Relationships
User	id, username, email, password, is_active	Django built-in authentication model.
Profile	user (FK), role (admin/customer/trainer)	Extends User with role permissions.
TrainerDetail	user (FK), category (gym/yoga/zumba)	Stores trainer specialization category.
Package	name, duration_months, price, description	Time-based membership plans (3m/6m/1y).
Membership	user (FK), package (FK), start_date, end_date	Active customer membership subscription.
Product	name, price, image, description	E-commerce supplement store items.
Payment	user (FK), name, plan, amount, card, transaction_type	Payment transactions and receipt records.
Client	trainer (FK), client (FK), goal, status, joined_on	Trainer-to-customer allocation mapping.
WorkoutPlan	trainer (FK), client (FK), title, description, duration	Assigned exercise workout routines.
DietPlan	trainer (FK), client (FK), breakfast, lunch, dinner	Assigned daily meal nutrition plans.
Schedule	trainer (FK), client (FK), session_date, session_time	1-on-1 scheduled coaching sessions.
Progress	trainer (FK), client (FK), weight, height, bmi, notes	Trainer-logged progress records.
Measurement	customer (FK), weight, height, bmi, month	Monthly customer BMI tracking log.
Feedback	customer (FK), rating (1-5), message, created_at	Customer star rating & review feedback.
Notice	subject, message, recipient_role, email_sent	Broadcast announcements & email logs.

4. UI/UX Design Engine & 60:30:10 Color Architecture

FitMaster AI implements a high-energy cyber visual theme structured around the professional **60:30:10 Color Rule**:

- **60% Dominant Base Background (#070e17)**: Deep midnight canvas providing optimal visual dark-mode contrast.
- **30% Structural Container Surface (#0f1d2e)**: Glassmorphism card surfaces, sidebar drawers, topbars, and table panels.
- **10% Focal Action Accents (#00ff87, #60efff, #58a6ff, #a855f7)**: Electric highlights used strictly for CTA buttons, active state indicators, key metric numbers, and chart curves.
- **Interactive Features**: High-Energy Motivation Marquee Ticker, Web Audio API Synth Sound Feedback (energy_audio.js), Three.js 3D Muscle Visualizer, Chart.js Analytics Graphs.

5. Installation & Command Execution Guide

1. Virtual Environment Activation:

```
.\.env\Scripts\Activate.ps1
```

2. Perform Database Migrations:

```
.\.env\Scripts\python.exe manage.py makemigrations  
.\.env\Scripts\python.exe manage.py migrate
```

3. Launch Development Server:

```
.\.env\Scripts\python.exe manage.py runserver
```

4. Verify Static Collection:

```
.\.env\Scripts\python.exe manage.py collectstatic --noinput
```